

National Institute of Technology Calicut

Department of Computer Science and Engineering

B. Tech | Course Plan (tentative) | Monsoon 2025-2026

CS4061D Topics in Data Analytics

Course Outcomes:

CO1 Apply various statistical inference techniques.

CO2 Apply and contrast different supervised and unsupervised machine learning techniques.

CO3 Apply data analytics techniques in the context of big/massive data.

Lecture Details: Lecture Hours: H slot | Lecture Room: MB-207

Instructor: Jayaraj P, Gopakumar G, B | Office: CSE 301-A, CSE 301-B |

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References

- T. Hastie, R. Tibshirani, and J. Friedman, The elements of statistical learning, 2e, Springer, 2009.
- D. C. Montgomery and G. C. Runger, Applied statistics and probability for engineers, John Wiley & Sons, 2010
- K. P. Murphy, Machine Learning, a probabilistic perspective, MIT Press Cambridge, Massachusetts, 2012
- A. Smola and S. V. N. Vishwanathan, Introduction to Machine Learning, Cambridge University Press, 2008.
- J. E. Hopcroft and R. Kannan, Foundations of Data Science, e-book, 2013

Grading and Mark distribution

Grading will be relative

Following is the split up of 100 marks

Midterm Test: 20 marks | Assignment: 20 marks | Quiz: 20 marks | End

Exam: 40 marks

All issues regarding valuation of exams must be resolved within one week after the marks are announced.

Absence for exams/assignments/quizzes without prior permission from the instructor will be equivalent to zero marks in the corresponding exam/assignment/quiz.

There will be no makeup exam except in case of genuine reason. In the event of such exceptional cases, the student must discuss the matter with the instructor and must get permission at least one day before the date of exam.

There will not be any change in the submission date once announced. Late submission/s will not be valued at any cost, and will be awarded zero marks.

Attendance

80% attendance is compulsory for writing the end semester exam.

Standard of Conduct

Each student is expected to adhere to high standards of ethical conduct, especially those related to cheating and plagiarism. Any submitted work MUST BE an individual effort. Any academic dishonesty will result in zero marks in the corresponding exam or evaluation and will be reported to the department council for record keeping and for permission to assign F grade in the course. The department policy on academic integrity can be found at <http://cse.nitc.ac.in/sites/default/files/Academic-Integrity.pdf>

Tentative Schedule

Week 01 Introduction to Data Analytics- Revision of Mathematical Basics - Probability, Statistics

Week 02 Linear Algebra and Calculus

Week 03 Data Preparation and Cleansing - Descriptive Statistics

Week 04 Probability Distributions- Inferential Statistics through hypothesis test

Week 05 Regression and ANOVA

Week 06 Machine Learning: Introduction and Concepts- Differentiating algorithmic and model based frameworks -

Week 07 Regression: Ordinary Least Squares, Ridge Regression, Lasso Regression, K Nearest Neighbours

Week 08 Midterm Test

Week 09 Regression Classification- Supervised Learning with Regression and Classification techniques -Bias- Variance Dichotomy

Week 10 Model Validation Approaches, Logistic Regression

Week 11 Linear Discriminant Analysis, Quadratic Discriminant Analysis, Regression and Classification Trees

Week 12 Support Vector Machines, Ensemble Methods: Random Forest , Neural Networks

Week 13 Deep learning, Unsupervised Learning and Challenges for Big Data Analytics – Clustering Associative Rule Mining

Week 14 Challenges for big data analytics Prescriptive analytics - Creating data for analytics through designed experiments

Week 15 Creating data for analytics through Active learning, Creating data for analytics through reinforcement learning

Week 16 Massive Data Analytics - Social Network Analytics - Security and Privacy Issues